Non-Traditional Composite/Operation Limits

The commonly memorized limit of composition rule generally is presented as something like this:

$\lim_{x\to a} f\big(g(x)\big) = f\left(\lim_{x\to a} g(x)\right)$ provided that $\lim_{x\to a} g(x) = L$ and $f(x)$ is continuous at $x = L$.

This is useful, but like many $if - then$ statements it gets misread as an $if\ and\ only\ if$ statement. If the $f(x)$ is not continuous at $x = L$ or if the limit of $g(x)$ does not exist at $x = a$, then we cannot apply the property. It does not tell us anything conclusive about the existence of $\lim_{x\to a} f\big(g(x)\big)$.

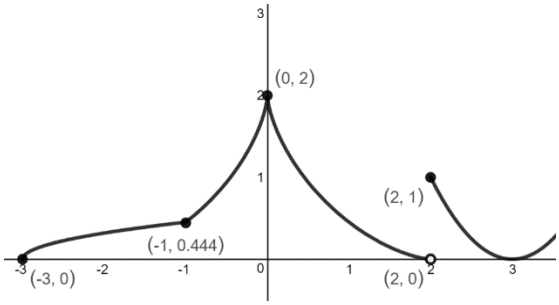
Similar discussions follow from all the 'standard' limit properties. E.g., If $\lim_{x\to a} f(x) = L_1$ and $\lim_{x\to a} g(x) = L_2$, then $\lim_{x\to a}(f(x) + g(x)) = L_1 + L_2$. However, if $\lim_{x\to a} f(x)$and/or $\lim_{x\to a} g(x)$ don't exist, we cannot draw any conclusion about $\lim_{x\to a}(f(x) + g(x)) = L_1 + L_2$.

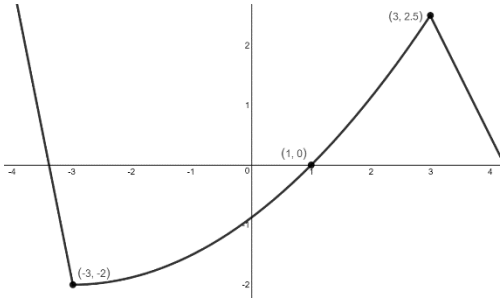
What can we do when the traditional limit properties don't apply? Well they also can be applied to one-sided limits. The most direct and effective route to exploring the limit is to remember another property of limits that IS an $if\ and\ only\ if$ statement:

$$\text{Iff } \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L, \text{ then } \lim_{x\to a} f(x) = L.$$

By exploring the right-hand and left-hand compositions or operations, we can often find if the limit being explored exists or not.

Consider: the graph of $f(x)$ the graph of $g(x)$



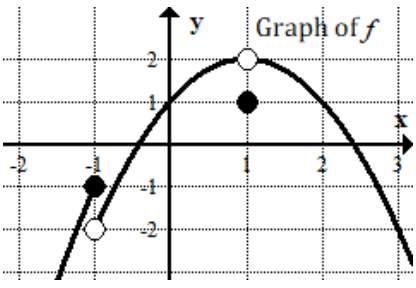


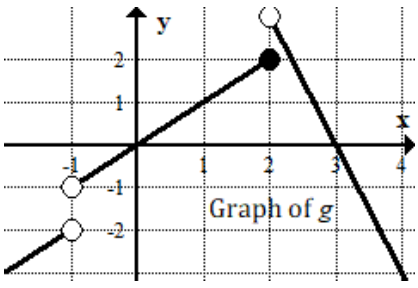
Find (a) $\lim_{x\to 1} f(g(x))$ (b) $\lim_{x\to 2}[f(x)\cdot g(x)]$ (c) $\lim_{x\to 0} f(f(x))$

(a) For this composition, traditional rules apply.

(b) While the limit of $f(x)$ doesn't exist at $x = 2$, explore the left-hand and right-hand limits of the products.

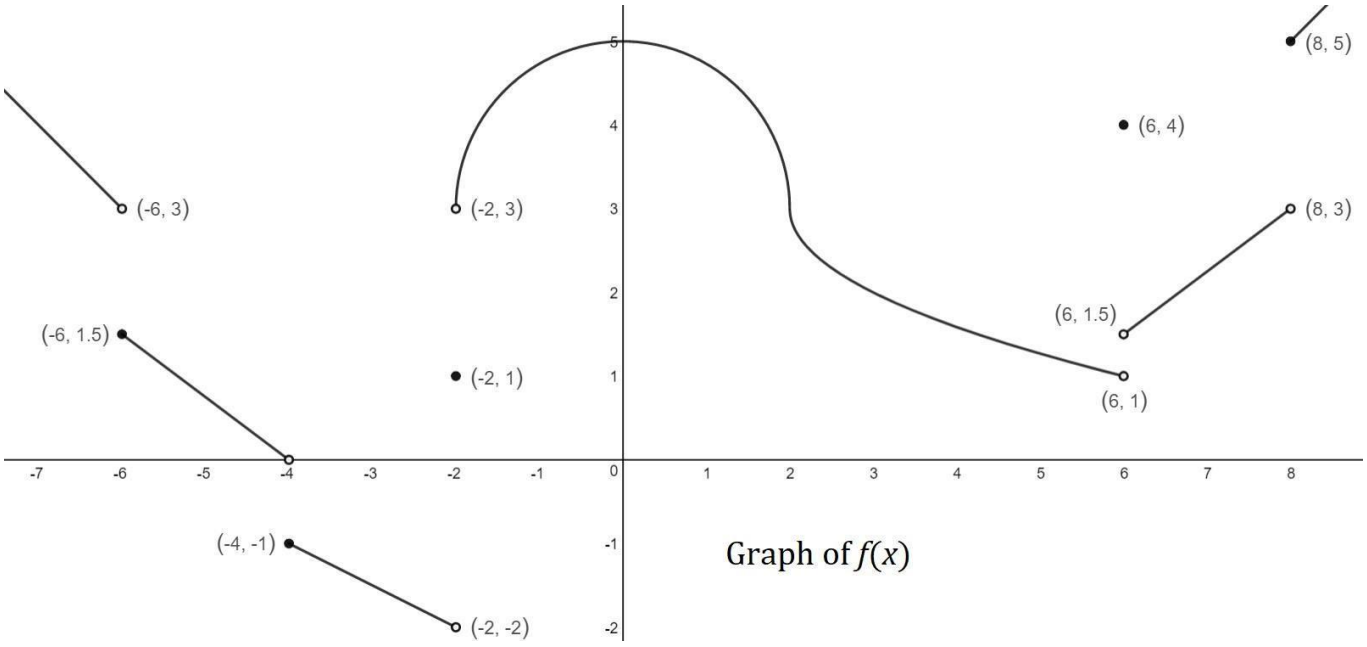
(c) It is important to consider from what direction the inner limit value is approaching 2. [Hint: Sometimes a tabular approach helps see what's happening, even if you don't know the actual function.]



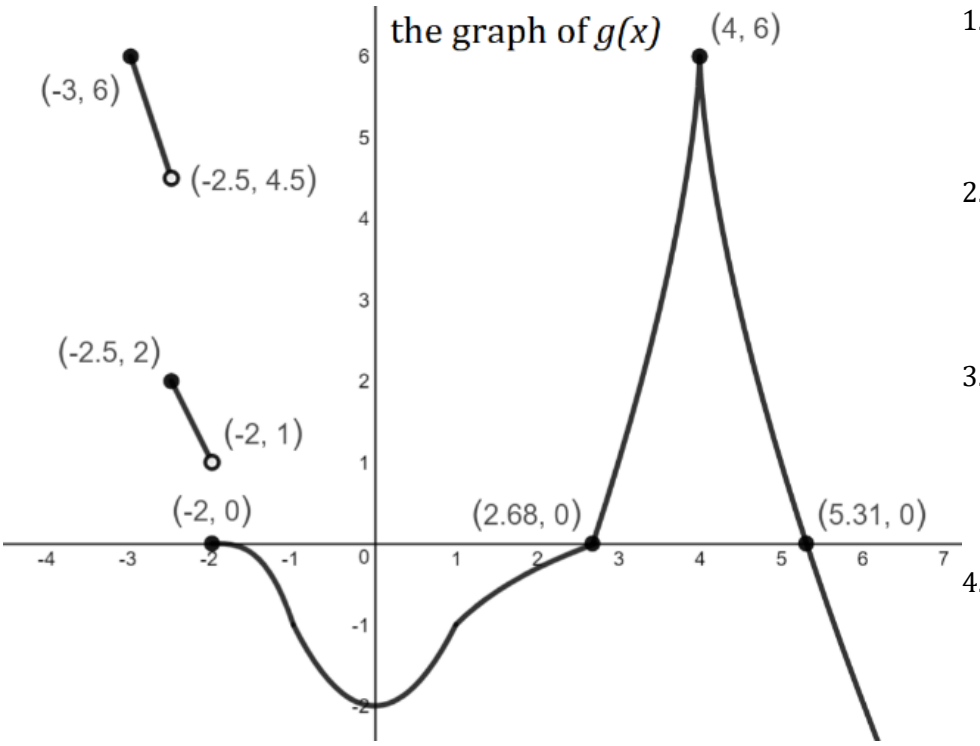


Find $\lim_{x\to -1}[f(x) + g(x)]$.

Find $\lim_{x\to 1}[f(x) + g(x+1)]$.



Graph of $f(x)$



1. Find $f(g(4))$.

2. Find $\lim_{x\to 4} f(g(x))$.

3. Find $f(g(0))$.

4. Find $\lim_{x\to 0} f(g(x))$.

5. Find $\lim_{x\to 6^-} g(1-f(x))$.

6. Find $\lim_{x\to 3^+} f(15-x^2)$.

7. Find $\lim_{x\to 2^-} g(2-x^2)$.

8. Find $\lim_{x\to -2^-} f(f(x))$.